HCI STUDENT ACTIVITY

Redesign a Messy Interface

Human perspective • Usability • Evolving technologies

Suggested duration	60–90 minutes

1. Activity challenge

You are a UI/UX design team hired by a university to improve its Student Portal. Students report difficulty finding courses, checking deadlines, registering for examinations, and accessing important messages. Analyse the current interface from a human-centred perspective and redesign it to be usable, accessible, inclusive, and appropriate for modern devices and technologies.

Design goal: improve how people perceive information, direct attention, remember actions, learn the system, and interact with it—not just how the interface looks.

2. Part A — Investigate the messy interface

In groups of 3–4, inspect the supplied messy Student Portal screenshot. Identify at least six usability problems.

No.	Problem observed	Why it affects users
1		
2		
3		
4		
5		
6		

Discuss:

- Which elements compete for the user's attention?
- What information should have the highest visual priority?
- How might the interface cause errors or frustration?
- How might a first-year student, a person with a visual impairment, and a student using a phone experience it differently?

3. Part B — Apply HCI principles from the human perspective

Human perspective / principle	Possible redesign decision
Perception	Readable typography, meaningful icons, sufficient contrast, clear visual hierarchy.
Attention	Highlight urgent deadlines and reduce unnecessary distractions.

Memory	Use familiar labels, visible navigation and reminders.
Learning	Make navigation predictable and provide helpful guidance.
Motor behaviour	Provide adequately sized touch targets and comfortable spacing.
Feedback	Confirm successful actions and show progress indicators.
Error prevention and recovery	Validate forms, explain errors and allow users to correct mistakes.
Accessibility and inclusion	Support keyboard navigation, screen readers, alt text and non-colour indicators.
User control	Allow users to review actions, change preferences and undo reversible actions.

Your task: choose at least five principles. Explain how each one motivates a specific change in your redesigned interface.

Principle	Change in our design	Expected benefit to the user

4. Part C — Integrate evolving HCI technologies

- **Responsive and mobile-first design:** Reorganise navigation for smaller screens; support touch interaction and different devices.
- **AI assistant / conversational interface:** Help users find deadlines or course information. Include limitations, privacy protections and access to human support.
- **Voice interaction:** Support selected tasks through speech while keeping visible alternatives.
- **Personalisation and context-aware support:** Offer configurable reminders and relevant information without overwhelming users with notifications.
- **Other emerging technology:** Propose a technology only if it addresses a real user need and can be justified.

Select at least two technologies and complete the table.

Technology	User problem addressed	Expected benefit	Risk / limitation and mitigation

5. Part D — Recreate the interface

Use paper sketches, Figma, Canva, PowerPoint, or a simple HTML/CSS prototype. Your redesign should include:

- A clear dashboard with logical information hierarchy.
- Consistent navigation and understandable labels.
- Course and assignment overview.
- Upcoming deadlines and timetable.
- Notifications with appropriate priority.
- Clear feedback and error recovery.
- Responsive and accessible interaction.
- At least two justified applications of evolving HCI technologies.

Design planning notes:

Target users and context: _____

Platform (web / mobile / multimodal): _____

Key HCI principles: _____

Chosen technologies: _____

Main screen layout and navigation: _____

6. Part E — Evaluate the redesign

Ask another group to test the prototype. Observe whether they can complete these tasks without help.

☐ Find the next assignment deadline.

☐ Locate the timetable.

☐ Find the course registration page.

☐ Find an unread message.

☐ Change a notification or accessibility preference.

Task / observation	What the tester did or struggled with	Change we would make

Discussion: Has the redesign actually improved the experience, or has it merely improved the appearance? Explain using evidence from testing.

7. Knowledge reflection (individual)

1. Which human capability or limitation most influenced your redesign, and why?

_____

_____

2. Which HCI principle had the greatest effect on usability? Give one example from your prototype.

_____

_____

3. How did your chosen emerging technologies improve interaction? What risks did they introduce?

_____

_____

4. What did user testing reveal that your team had not anticipated?

_____

_____

5. If you redesigned the interface again, what would you change and why?

_____

_____

8. Expected learning outcomes

1. Analyse interfaces using human perception, attention, memory, learning and motor behaviour.
2. Apply established HCI principles to redesign an existing interface.
3. Evaluate benefits and limitations of emerging interaction technologies.
4. Design interfaces that accommodate diverse users and contexts.
5. Justify design decisions using user feedback and reflective learning.

Lecturer's closing question

If an interface uses the latest AI technology but makes users confused, anxious or dependent, can it still be considered good HCI design? Why or why not?